

Experiment 2: Bubble Sort – Traffic Signal Priority Queue

Aim

To implement the **Bubble Sort** algorithm in Python to arrange vehicles according to their priority in a traffic signal queue.

Problem Statement

At a traffic signal, vehicle priorities are:

Ambulance > Bus > Car

Whenever a new vehicle enters the queue, reorder the queue using Bubble Sort based on priority.

Algorithm

Step 1: Start.

Step 2: Read the number of vehicles.

Step 3: Read the vehicle names.

Step 4: Assign priorities:

- Ambulance = 1
- Bus = 2
- Car = 3

Step 5: Compare adjacent vehicles.

Step 6: Swap them if their priorities are incorrect.

Step 7: Repeat until the queue is sorted.

Step 8: Display the priority queue.

Step 9: Stop.

Source Code (Python)

main.py



```
1 # Bubble Sort - Traffic Signal Priority Queue
2
3 priority = {
4     "ambulance": 1,
5     "bus": 2,
6     "car": 3
7 }
8
9 n = int(input("Enter number of vehicles: "))
10
11 queue = []
12
13 print("Enter vehicles (ambulance/bus/car):")
14 for i in range(n):
15     queue.append(input().lower())
16
17 for i in range(n - 1):
18     for j in range(n - i - 1):
19         if priority[queue[j]] > priority[queue[j + 1]]:
20             queue[j], queue[j + 1] = queue[j + 1], queue[j]
21
22 print("Priority Queue:", queue)
```

Output

Output

```
Enter number of vehicles: 4
Enter vehicles (ambulance/bus/car):
car
car
bus
ambulance
Priority Queue: ['ambulance', 'bus', 'car', 'car']
```

=== Code Execution Successful ===

Complexity Analysis

Aspect	Complexity
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Time Complexity	$O(n^2)$
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Space Complexity	$O(1)$
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Result

The Bubble Sort algorithm was successfully implemented to arrange vehicles according to their priority. The program correctly reordered the queue so that Ambulance is first, followed by Bus and then Car.